Demand Forecasting and Inventory Optimization

A case study using the Online Retail II dataset

Production and Industrial Engineering

Ukkusuri Bharath | Roll No: 124107009

Department of Production and Industrial Engineering, National Institute of Technology Kurukshetra

1. Summary

A retail shop must decide how much of each product to expect to sell and how much to order, and when. This project answers both questions using two years of real transaction data. Weekly demand for the ten highest-value products was forecast with two methods, products were ranked by sales value with ABC analysis, and order quantities, safety stock and reorder points were calculated for the top ten products.

Main findings: 1,052 of 4,870 products (22%) generate 80% of sales value. Exponential smoothing with a small smoothing factor (alpha 0.1) was the most accurate method for 7 of the 10 products, the 3-week moving average for 2 products, and alpha 0.5 for the seasonal Christmas item. Economic order quantities range from about 711 to 5,096 units, giving 8 to 18 orders per year.

2. Problem Statement

The shop owner needs to answer two questions: (1) how much of each product will we sell next week, and (2) how much should we order, and at what stock level should we reorder? Ordering too little causes lost sales, and ordering too much ties up money in stock.

3. Data

The Online Retail II dataset contains invoice-level sales of a UK online giftware retailer from December 2009 to November 2011. The raw data had 1,067,371 rows. After cleaning, 1,008,690 valid rows remained.

Cleaning steps:

- Removed 22,523 duplicate rows caused by overlap between the two source sheets.
- Removed cancellations, zero or negative quantities and prices, and non-product codes such as postage and bank charges.
- Removed 6,255 orders that were later cancelled in full, which had created false demand (single orders of about 74,000 and 80,000 units).
- Kept only complete Monday-to-Sunday weeks. Weeks 4 (28 Dec 2009) and 56 (27 Dec 2010) are zero for every product because the company was closed.

Two files were prepared: a weekly demand table (104 weeks) for the ten highest-value products, and a product table of 4,870 products with price, annual demand and annual value.

4. Methodology

4.1 Demand forecasting

Two methods were compared for each product. The 3-week moving average forecasts next week as the mean of the last three weeks. Exponential smoothing forecasts next week as alpha x last actual + (1 - alpha) x last forecast, and three values were tested (alpha = 0.1, 0.3, 0.5). Accuracy was measured by Mean Absolute Deviation (MAD) on the last 12 weeks, so every method was judged on the same period. The method with the lowest MAD was chosen for each product.

4.2 ABC analysis

Products were sorted by annual sales value (price x annual demand). A cumulative share of total value was calculated. Products up to 80% were classed A, up to 95% class B, and the rest class C.

4.3 Economic order quantity (EOQ)

EOQ = sqrt(2 x D x S / H), where D is yearly demand, S is the cost of placing an order and H is the holding cost per unit per year. It gives the order size that minimises the total of ordering and holding cost.

4.4 Safety stock and reorder point

Safety stock = z x sigma x sqrt(L), where sigma is the standard deviation of weekly demand and L is the lead time in weeks. Reorder point = average weekly demand x L + safety stock.

4.5 Assumptions

The dataset contains no cost or supplier information, so the following were assumed:

- Ordering cost: 50 GBP per order.
- Holding cost: 20% of unit price per year.
- Lead time: 2 weeks.
- Service level: 95% (z = 1.65).

5. Results

5.1 ABC analysis

Class	Products	Share of products	Share of sales value
A	1,052	22%	80%
B	1,275	26%	Next 15%
C	2,543	52%	Last 5%

About one fifth of the products generate four fifths of the sales value, so inventory effort should focus on class A. All ten products studied here are class A.

5.2 Forecast accuracy (MAD, last 12 weeks, units per week)

Product	Moving avg	alpha 0.1	alpha 0.3	alpha 0.5	Best method
Regency Cakestand	64.0	57.7	62.5	70.8	alpha 0.1
White Hanging Heart T-Light Holder	322.9	292.3	310.9	360.6	alpha 0.1
Jumbo Bag Red Retrospot	687.8	481.9	558.2	595.8	alpha 0.1

Product	Moving avg	alpha 0.1	alpha 0.3	alpha 0.5	Best method
Party Bunting	88.5	186.1	103.8	95.9	Moving average
Assorted Colour Bird Ornament	297.9	235.7	237.4	246.0	alpha 0.1
Paper Chain Kit 50's Christmas	343.0	683.8	375.2	301.6	alpha 0.5
Chilli Lights	173.8	167.7	169.4	181.4	alpha 0.1
Small Popcorn Holder	555.7	906.8	698.2	646.3	Moving average
Jumbo Bag Pink Polkadot	265.3	262.6	271.5	287.0	alpha 0.1
Lunch Bag Red Retrospot	130.1	114.9	120.4	125.1	alpha 0.1

Exponential smoothing with alpha 0.1 was best for 7 products. Their weekly sales fluctuate sharply, and a small alpha avoids chasing every spike. Party Bunting and Small Popcorn Holder did better with the moving average, which reacts faster to their large swings. Paper Chain Kit is a Christmas item with a strong seasonal pattern, so a faster alpha (0.5) fitted it best.

The forecast error is between roughly 26% and 57% of average weekly sales. This is large, but typical for lumpy wholesale-type demand, and it is the reason safety stock is needed.

5.3 Order quantity, safety stock and reorder point

Product	Annual demand	EOQ	Orders / year	Safety stock	Reorder point
Regency Cakestand	12,894	711	18.1	416	905
White Hanging Heart T-Light Holder	45,480	2,776	16.4	1,158	2,860
Jumbo Bag Red Retrospot	48,012	3,509	13.7	1,106	2,921
Party Bunting	14,008	1,190	11.8	562	1,098
Assorted Colour Bird Ornament	39,922	3,437	11.6	1,621	3,105
Paper Chain Kit 50's Christmas	17,330	1,714	10.1	1,220	1,818
Chilli Lights	7,906	894	8.8	445	741
Small Popcorn Holder	44,152	5,096	8.7	2,107	3,707
Jumbo Bag Pink Polkadot	19,506	2,236	8.7	593	1,342
Lunch Bag Red Retrospot	19,968	2,460	8.1	410	1,167

Example: for the Regency Cakestand, order about 711 units each time (roughly 18 orders a year, or one every three weeks). When stock falls to about 905 units, place the next order. Of these, about 489 units cover expected sales during the 2-week delivery wait and about 416 units are safety stock.

6. Recommendations

- Concentrate stock control on class A products, which carry 80% of the sales value.
- Use exponential smoothing with alpha 0.1 as the default forecast, but use the 3-week moving average for Party Bunting and Small Popcorn Holder.
- Use a faster alpha (0.5) for seasonal items such as Paper Chain Kit, and review them before the Christmas season.
- Order in the EOQ quantities shown and reorder when stock reaches the reorder point.

- Keep the higher safety stock for products with erratic demand (Small Popcorn Holder, Assorted Colour Bird Ornament), and review it regularly.

7. Limitations

- Order cost, holding cost, lead time and service level are assumed, not taken from the data. Real values should replace them in practice.
- Demand is gross sales. Partial returns were not subtracted.
- Prices are selling prices, not purchase costs.
- Forecasts were tested on only the last 12 weeks.
- Some products have lumpy or seasonal demand (Paper Chain Kit has 22 zero weeks), so errors are large.
- EOQ, safety stock and reorder points were calculated for the top ten products only. The same method extends to all 1,052 class A products.

8. Conclusion

The project shows a complete demand-to-inventory workflow on real data: forecasting, ABC classification, EOQ, safety stock and reorder point. It converts raw sales history into specific decisions for each product. The main next step is to replace the assumed costs and lead time with real company figures and to extend the analysis to every class A product.

9. Tools

Google Sheets (formulas, forecasting, ABC analysis, EOQ, safety stock and reorder point). Data source: Online Retail II dataset (UCI Machine Learning Repository).